Date:

YOUVA

109123 MATHS

$36 = 2^2 \times 3^2$
$45 = 3^2 \times 5$

1. $LCM(36, 45) = 2^2 \times 3^2 \times 5 = 180$ sec $= 3$ min

$36 = 2^2 \times 3^2$

$45 = 3^2 \times 5$

$\boxed{(3)\ 3}$

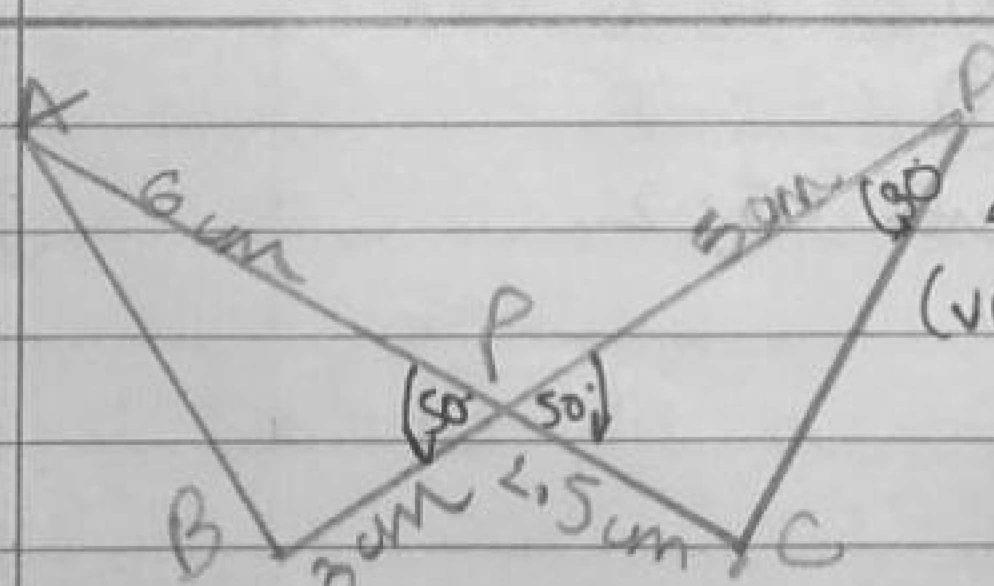
2. $\dfrac{2\tan 30^\circ}{1-\tan^2 30^\circ} \Rightarrow \dfrac{2\left(\frac{1}{\sqrt{3}}\right)}{1-\left(\frac{1}{\sqrt{3}}\right)^2} = \dfrac{\frac{2}{\sqrt{3}}}{\frac{3-1}{3}}$

$= \dfrac{2/\sqrt{3}}{2/3}$

$= \dfrac{\frac{1}{\sqrt{3}}}{\frac{1}{3}}$

$= \dfrac{3}{\sqrt{3}} = \sqrt{3}$

$\boxed{(3)\ \tan 60^\circ}$

3 

$\angle APB = \angle CPD = 50^\circ$
(vertically opposite angles)

$\dfrac{PA}{PD} = \dfrac{PB}{PC} = \dfrac{6}{5} = \dfrac{3}{2.5} = 1.2$

by SAS similarity $\Delta APB \sim \Delta DPC$

$\angle PAB = \angle PDC = 30^\circ$

$\angle PBA = 180 - (50 + 30^\circ) = 180 - 80^\circ = 100^\circ$ $\boxed{(4)\ 100^\circ}$

12 HCF $(616, 32) = 2^3 = 8$

$616 = 2^3 \times 7 \times 11$

$32 = 2^5$

2	616
2	308
2	154
7	77
11	11
	1

$\therefore$ 8 rows

$\boxed{(1)\ 8}$

13 $\boxed{\text{(1) similar}}$

14. $a = x^3y^4z^2$

$b = x^2yz^4$

$LCM(a, b) = x^3y^4z^4$

$\boxed{(1)\ x^3y^4z^4}$

15. $\alpha = (-3)$

$\beta = (4)$

$\alpha+\beta = -3+4 = 1$

$\alpha\beta = (-3)4 = (-12)$

$k[x^2 - x(\alpha+\beta) + \alpha\beta]$.

$k[x^2 - x - 12]$.

$k\left[\frac{x^2}{2} - \frac{x}{2} - 6\right]$

ans : $\boxed{(3)\ \frac{x^2}{2} - \frac{x}{2} - 6}$

16 $\boxed{\text{(3) intersecting or coincident}}$

17 $\boxed{\text{(1) parallel.}}$

18 let $\Delta ABC \sim \Delta PQR$

$\frac{ar(ABC)}{ar(PQR)} = \frac{AB^2}{PQ^2} = \frac{(4)^2}{(9)^2} = \frac{16}{81}$

$\boxed{(4)\ 16:81}$

19 A :

$x^2 - 4x - 5$

$x^2 - 5x + x - 5 = 0$

$x(x-5) + (x-5) = 0$

$(x+1)(x-5) = 0$

$x = 5; (-1)$

R: $\alpha = 2+\sqrt{3}$

$\beta = 2-\sqrt{3}$

$\alpha+\beta = 4$

$\alpha\beta = (4-3) = 1$.

$p(x) = x^2 - 4x + 1$.

$\boxed{\text{(3) Assertion (A) is true but Reason R is false.}}$